# Andrew Pols

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## Education

**University of Toronto** Sep. 2024 - Aug. 2028

- HBsc., **Computer Science and Statistics** | **GPA: 4.00 / 4.00**
- Relevant coursework: Software Design; Computer Organization; Software Tools and Systems Programming

## Experience

**Software Developer Intern** May 2026 - May 2027

*IBM Canada Ltd.*

- Built a dependency graph for IBM Planning Analytics Workspace in React and TypeScript, helping clients trace their financial models while staying responsive on 2,500+ objects by rendering only the relevant subset.
- Achieved fast loading by traversing relationships with lazy-loading, caching, and concurrent asynchronous requests to TM1 REST APIs, cutting a manual trace across thousands of objects from hours to minutes.
- Worked with design, product management, QA, and engineering in Agile sprints to define graph layout, filtering, and object-specific interactions for large dependency structures, then shipped the feature to clients.

## Projects

**ScholarSearch** | *Source Code* **ReactJS | Python | PostgreSQL | FastAPI**

- Built a research paper search engine that indexes 1,000,000+ papers using PostgreSQL for persistence and ChromaDB for vector embeddings, returning scalable semantic search in under 200ms with Redis caching.
- Built D3.js citation graphs so researchers can see a paper's impact and related work in an interactive page.
- Containerized Postgres, ChromaDB, and Redis via Docker giving our team a consistent development setup.

**FindMySound** | *Source Code* **ReactJS | Python | Django | Scikit-Learn**

- Built a web app that recommends music based on a user's music library, using Spotify's Web API and OAuth 2.0.
- Applied a KNN model on ingested BERT feature vectors to recommend music, reaching 95% user satisfaction.
- Handled user accounts through a custom Django user model with JWT-based authentication for secure sessions.
- Exposed RESTful endpoints for authentication, account syncing, and recommendations for frontend features.

**GroupFlow** | *Source Code* **Java | MongoDB | Swing**

- Built a Java desktop app giving university students a central place to schedule meetings and assign tasks.
- Used MongoDB to persist shared data of users, tasks, and meetings, with Java Swing for the desktop interface.
- Followed Clean Architecture and SOLID principles by separating each use case into its own isolated package.
- Collaborated in a team of 6, enforcing CheckStyle and requiring thorough PR reviews before merging code.

## Technical Skills

- **Programming Languages**: Java, Python, C, JavaScript, TypeScript, SQL, HTML5, CSS
- **Frameworks & Libraries**: React, Node.js, Next.js, Django REST, Flask, React Flow, Scikit-learn, OpenCV
- **Developer Tools**: AWS, Git, Jira, Docker, Azure, Redis, PostgreSQL, MongoDB, Nginx, UNIX